Critical Components and System-Level Resilience in Professional Sports Teams: A Literature Review

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Abstract

This review examines the concept of the "critical component", a structural or functional part of a system whose removal disproportionately reduces overall performance, and applies this framework to professional sports teams. Based on literature from network science, organizational and health-system resilience, family-business studies, and sports science, the review defines four recurring concepts - critical component, redundancy, shock, and recovery - and compares each stage's similarities and differences across various systems. The review identifies a gap in the existing literature: most studies treat player injuries or losses as equivalent, interchangeable events rather than examining how the pre-loss importance and role of the missing component shape the magnitude and duration of a team's decline. The review concludes by proposing future research using two decades of National Hockey League (NHL) data intended to address this gap.

I. Introduction

Teams in professional sports are complex adaptive systems where players are part of the system and collaborate in their different roles to reach a shared goal [8]. As in any complex system, such as a biological network, a healthcare system, an ecosystem, or a computer network, the functioning of the whole relies more on how the individual parts interact than on the actions of any one part [1], [3]. When an important player is injured, suspended, retires, or changes teams, the loss may disrupt the entire system.

The level of disruption caused by the loss, however, is determined by several factors, including the importance of the missing component, the level of redundancy in the system, and how well the system is able to bounce back from the shock. These concepts have been extensively researched in other areas of life and offer a theoretical framework that can be applied to team performance.

In resilience literature, a shock is defined as an unusual event that interrupts the normal functioning of a system, while recovery describes the process of returning to or exceeding its previous level of functioning [2], [16]. Whether a health system, a business organization, an ecological community, or an engineering network, shocks have the potential to cause performance issues, and the most resilient systems have properties that enable them to operate even if some of their parts fail. Applied to professional sports, these concepts may help explain why some teams recover quickly after losing a key player, while others experience longer declines.

II. Research Question and Key Concepts

It is helpful to define four terms that appear throughout the literature on resilience: critical component, redundancy, shock, and recovery. These terms help explain why some losses are more significant than others and how systems respond to loss. Together, they frame the central question addressed by this review: How does the loss of a critical component affect the performance and recovery of a system such as a professional sports team?

Sources published in English through June 2026 were included if they examined how the loss or failure of an important component affected the performance or recovery of a larger system. Research from sports science, healthcare, business, and ecology was included when its concepts could be applied to professional sports teams.

A critical component is one whose removal leads to the overall function of the system becoming significantly reduced relative to the loss of some other component(s). In other words, a critical component is a structural or functional piece whose failure results in a greater decrease in system performance than the failure of any other component. In engineering and network science, nodes that, if removed, significantly affect the functioning of the system due to the many important links between them are called critical components [1]. Likewise, in biological systems, the loss of some genes, proteins, or species disrupts the function of the system as a whole [1]. This concept is also applicable in other areas of life outside of sport. Hynes et al. [2] suggest that a highly optimized supply chain and health system may be vulnerable if efficiency is achieved at the expense of spare capacity.

Redundancy, on the other hand, measures how much a system has interchangeable or back-up capacity to absorb the failure of one of its components without a corresponding decrease in system performance. At the level of national economies and supply chains, Hynes et al. [2] make this point, noting that many global supply chains and health systems are very efficient and tightly optimized, and that this comes at a cost of reduced redundancy, reducing their resilience to shocks such as the COVID-19 pandemic. Health-system resilience also depends on maintaining critical services and resilience across system elements and public-health capacities [16]. The latest health-system resilience synthesis suggests resilient systems keep critical functions in place by combining a level of redundancy with critical resources. A resilient system is not reliant on a single element, but relies on several interdependent elements that can make up for the failure of one.

These concepts can be applied directly to any professional sports team. All athletes play a part in a team's performance, but some are more influential - through leadership, creativity, or skill - than others. These players are the most critical to the team, as their impact goes beyond their statistics to influence the organization as a whole. Based on their research, Calleja-González et al. [4] suggest that player availability is one of the most important factors determining success in elite team sports, as reduced availability is associated with poorer team performance and may limit substitutions and rotations. The more versatile a team is in terms of players' abilities, the more redundant it will be, making it easier to find replacements who can cover the same role well. If, however, certain players have very specific tactical and/or leadership requirements that cannot be easily filled, the effect of their absence becomes greater, and redundancy is reduced.

The same holds on a smaller scale for family businesses: Jayakumar and De Massis [5] argue that family firms with distributed leadership, informal capital and external networks may be better positioned to deal with shocks during the pandemic than those focused on a single decision maker.

A shock is an abrupt, typically unexpected, event that disrupts a system's normal state. Hynes et al. [2] view the COVID-19 pandemic as an exogenous shock to interconnected systems. Jayakumar and De Massis [5] expand this notion by describing how a single shock can turn into multiple compounding shocks - a "cocktail of shocks" - within interdependent subsystems, specifically the business, family and ownership subsystems of a family business. This multi-faceted view of shock is useful in sports, since the loss of a star player can cause a host of sub-shocks to ripple through a team, arising from shifts in roles, motivation and tactics.

Lastly, recovery describes the process - and the speed - with which a system returns to baseline levels or exceeds its pre-shock functioning. In their recent article on food systems during COVID-19, Klassen and Murphy [6] warn that a "recovery" that returns output to pre-existing levels without addressing the underlying inequities and vulnerabilities the shock exposed is not truly a recovery at all; to genuinely recover, systems must "build back better." Similarly, Hynes et al. [2] argue that resilient recovery should not just be about getting back to where a system was, but about "bouncing forward" to a better, less vulnerable state. In sports, this may mean that recovery after the loss of one or more significant team members should not be judged solely by whether results return to pre-loss levels, but also by whether the team becomes more resilient going forward.

III. Review of the Literature

This is a narrative rather than a systematic review: rather than applying a single formal search protocol, it draws together and synthesizes findings from sports science, network science, organizational and public-health resilience research, and family-business studies to build a cross-domain picture of how systems respond to the loss of a critical component. The findings are organized below around three guiding questions.

A. What Is a Critical Component?

Rather than all players contributing equally to team performance, a growing body of literature focuses on the idea that some players may be more structurally important than others. One of the clearest illustrations of this concept comes from Hägglund et al. [7]. After observing 24 European football clubs over 11 seasons, the authors found that greater injury burden and lower match availability were associated with poorer league rankings and European Cup performance. These findings show that player availability affects team outcome over time. However, because the study does not distinguish between the loss of central versus peripheral players, it does not establish whether the loss of a highly influential player causes a greater decline.

A complementary, more theoretical perspective is offered by Park [8], who uses a network model to examine how failures spread through interconnected sports systems and how highly connected components can influence overall performance. Applied to a professional team, this suggests that a key player does not necessarily have to be the most statistically productive, but rather the most influential in connecting other players, whether as a playmaker linking defense and attack or a captain who provides a base of support in times of stress. Together, Hagglund et al. [7] and Park [8] suggest that the loss of more central components may produce greater system-level disruption than the loss of peripheral ones.

B. What Happens When a Central Component Is Removed?

A related question is what changes once a central component is removed: how do remaining players take over the workload and responsibilities the missing player had carried? This effect is especially observable in the NBA, where in-game injuries and disqualification after six personal fouls can serve as unexpected, exogenous shocks that remove a player from active play and allow changes in teammates' performance to be observed, according to Kocsoy [9]. Because the change is sudden and unplanned, this design helps isolate the immediate effects of a leader's absence from broader, season-long trends. In particular, the study finds that the absence of players who are both team captains and All-Stars produces significant negative spillover effects on teammates' performance, while the absence of captain-only or All-Star-only players does not, supporting the idea that not all key players are equally indispensable.

This finding is consistent with Hägglund et al. [7], who found that greater injury burden and lower player availability were associated with poorer team performance. Both studies suggest that the mechanism behind the drop in performance is not simply the removal of one player's output, but a broader redistribution problem: teammates may take on new duties and roles, tactical formations built around the removed player may need to be reorganized, and the redistribution process itself may create inefficiencies that magnify the loss.

A similar pattern appears outside sports. In a business subsystem, a financial shock is unlikely to stay contained: it can create a parallel shock to the family subsystem when, for instance, a senior leader is unable to continue in their role, forcing an unprepared redistribution of authority within the family system [5]. At the level of national economies, a comparable argument applies - shocks in one system, such as public health, spread quickly to structurally linked systems such as trade, logistics, and employment [2]. In both contexts, the shock of losing a key component is not a single point-in-time event, but an ongoing problem that unfolds unevenly across an interconnected system, similar to the redistribution effects described in sports by Hägglund et al. [7] and Kocsoy [9].

The immediate issue is not just the missing player's individual output. Unfamiliar minutes, decisions, and tactical responsibilities may need to be assumed by the remaining players. Kocsoy [9] examines unexpected shocks in the form of injuries and foulouts by NBA players and finds that injuries to players who were both team captains and All-Stars negatively affected their teammates' productivity. However, the absence of a captain-only or All-Star-only player did not produce a statistically significant impact on teammates' performance. The discovery lends weight to the idea that formal leadership status does not necessarily mean that a player is critical. However, the combination of leadership and individual talent may lead to a stronger spillover effect at the team level.

The same study provides a quantitative example of the impact on outcomes. The probability of winning dropped by approximately 24% for home teams and 20% for away teams when a captain-All-Star was absent, after controls and fixed effects were included [9].

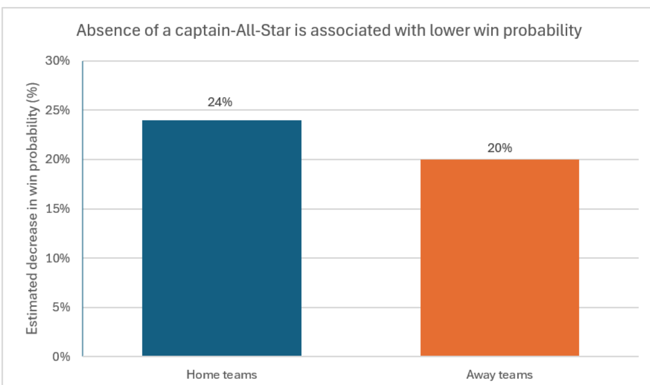

Figure 1. Estimated decrease in win probability associated with the absence of a captain–All-Star. Author-created visualization using adjusted estimates reported by Kocsoy [9, Table 7 and Table S6].

The disruption also affects decision-making. Kocsoy [9] found that two-point shots accounted for 74.4% of the combined two- and three-point attempts when a captain–All-Star was on the court, but this proportion dropped to 60.8% when the player was absent. The corresponding increase in 3-point shooting indicates that the affected team may rely on riskier or less familiar offensive strategies without a central organizer.

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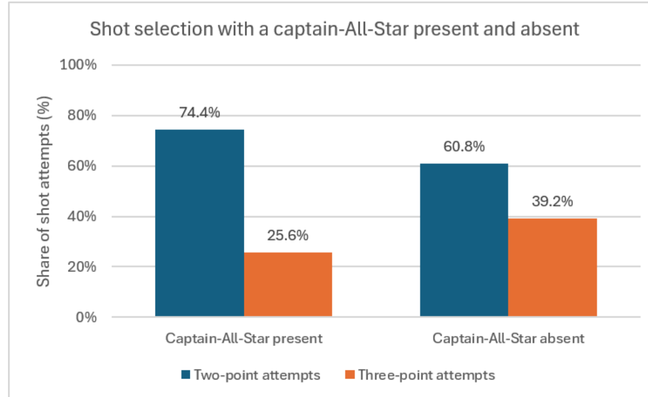

Figure 2. Shot-selection distribution with a captain-All-Star present versus absent. Author-created visualization using two-point attempt shares reported by Kocsoy [9]; three-point shares calculated as remainder of total two- and three-point attempts.

Research to date has shown that return-to-play protocols are not only beneficial for the athlete who suffered an injury, but also involve the wider sporting organization. Buchheit et al. [13] observed that professional football organizations employ progressive reintegration techniques, player workload monitoring, and player communication to help with reintegration and adaptation. The results suggest that the return-to-play process involves not only the injured player, but also the wider sporting structure. Likewise, studies in the area of performance analysis have revealed how the loss of key players often affects attacking and defensive formations, which then have to adjust to make the team more effective, before new tactical relationships are formed.

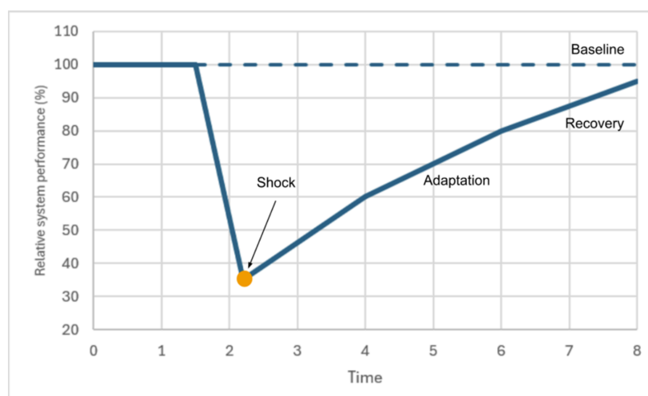

Figure 3. Author-created conceptual shock–recovery curve illustrating system performance following removal of a critical component, informed by the resilience framework of Artime et al. [12].

C. What Is Resilience?

In this context, resilience is defined as the ability of a system to continue functioning under stress and to return to a previous level of functioning - not merely to avoid disruption altogether. Using a dynamic network perspective on athletic performance, Den Hartigh et al. [10] suggest that as a system's resilience is depleted by repeated or intense stressors, the recovery time following each additional shock increases, and that this lengthening recovery time can be an early warning sign of future decline in performance. This offers a way of conceptualizing resilience as recovery, framing the key questions as how long recovery takes and whether the system shows measurable "improvement" after a shock.

Hurley [11] adds a psychological dimension to this picture through an opinion article discussing international rugby teams competing in the 2015 Rugby World Cup, several of which lost key players to tournament-ending injuries during the competition. The article suggests that the mental and emotional state of the rest of the team - confidence, cohesion, and understanding of new roles - may be an important factor in how quickly a team's on-field performance recovers, and that resilience may depend on how a team copes with the loss of one of its key parts. Den Hartigh et al. [10] similarly indicate that both structural network dynamics and the psychological adaptation of the team shape recovery curves after a shock.

The distinction between simply returning to baseline and truly recovering is explained clearly outside of sport, and it helps clarify what "recovery" should mean in this context. Klassen and Murphy [6] state that a food system that produced the same amount of food as before the pandemic, without changing the inequities the shock exposed, was not "recovered" - genuine recovery means fixing the vulnerabilities a shock exposed, not simply resuming pre-shock production. This is the difference between "bouncing back" to a preexisting, vulnerability-prone state and "bouncing forward" to a more resilient one [2]. Applied to sports, this means that if a team returns to its pre-loss level of play but remains just as dependent on a single irreplaceable player, it may have restored performance without fully addressing the fragility that made the loss so damaging in the first place. A fuller account of recovery, building on Den Hartigh et al. [10], would need to consider not only how quickly results rebound, but also how dependent the team's underlying structure remains on any single critical component going forward.

Applying these cross-domain concepts to professional sports may help explain why certain teams bounce back from missing a key player in the first year, while others cannot seem to get the same level of performance for several years afterward [15]. Resilience is the way a team copes with the loss of one of its important members. Commonly, recovery is gauged by the improvement of the match performance, league position, number of expected goals, player availability, or tactical efficiency in a number of games. While the focus is traditionally on outcomes at the end of the season, health-systems research also emphasizes recovery time as a measure of how long a system takes to return to its expected level after disruption [16]. Applied to sports, this suggests examining how long performance takes to return to the expected level after a player is lost. Recovery may take longer for teams that are less resilient due to their lack of tactical flexibility, smaller rosters, poor coaching, and lack of organizational support. Teams with little redundancy, on the other hand, may suffer a long decline as they are not able to redistribute tasks successfully. Resilience thus requires not only a team's response to disruption, but also its ability to adapt to changing circumstances over time.

Table 1. Qualitative Cross-Domain Comparison of shock, recovery, and redundancy across complex systems.

Domain	Shock	Recovery	Redundancy
Healthcare	Pandemic	Staff redeployment	High
Business	Financial crisis	Strategic change	Moderate
Network	Node failure	Backup routing	High
Sports	Player injury	Tactical adjustment	Moderate

Note: "High" indicates substantial backup capacity or multiple alternative pathways. "Moderate" indicates some replacement capacity but greater role limitations.

Another good example of resilience is found in health systems. Recent resilience frameworks refer to recovery as the ability to maintain critical services during a crisis and adapt to future challenges. Recovery is therefore a combination of keeping activity running in the short term and building long-term capacity [14]. Resilient systems are prepared, flexible, redundant, adaptive, and capable of learning, which allows them to recover better from future shocks.

IV. Evaluation of the Quality of the Literature

The literature reviewed varies in strength and methodology. Hagglund et al. [7] provide strong evidence over a long time interval by examining professional football clubs over 11 seasons. However, the design cannot confirm that injuries directly caused poorer performance or distinguish losses by player importance. Kocsoy [9] provides more direct evidence of spillovers by examining unplanned in-game losses, but the findings are limited to the NBA and define influential players through captaincy and All-Star status. Other sources, including opinion articles, theoretical models, and studies from healthcare and ecology, are useful frameworks but offer less direct evidence for sports teams.

It is clear from the studies reviewed above that the loss of a key component negatively affects a system's performance, and that some components matter more than others. Yet there is a gap in the literature when it comes to examining the link between the importance of a lost component and the magnitude and duration of the resulting decline in system performance. The current body of literature largely treats injuries or shocks as equivalent, interchangeable events rather than as individual events with their own characteristics. Further, the existing studies do not focus closely on the timing of shocks or on the pre-shock characteristics of the individual components involved.

V. Synthesis and Conclusion

Taken together, the literature reviewed here supports a consistent picture across domains: systems are more vulnerable to the loss of central, highly connected components than to the loss of peripheral ones, and true recovery requires addressing the structural dependence that made the loss damaging, not merely restoring prior output. What the literature has not yet done is connect the pre-shock importance of a specific component to the size and duration of the resulting decline. Future research could address this gap by gathering two decades of National Hockey League (NHL) competition data, linking pre-shock component performance with team context, and treating injuries as independent, system-wide organizational shocks.

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